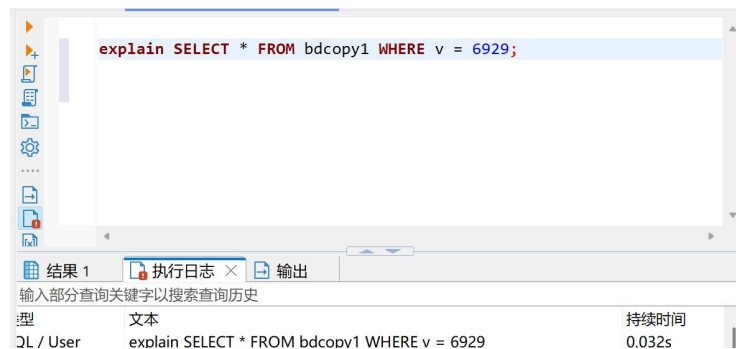


第四次实验

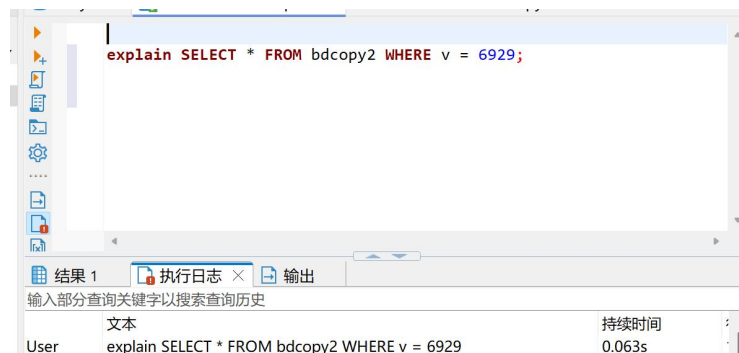
TASK 1

Q1

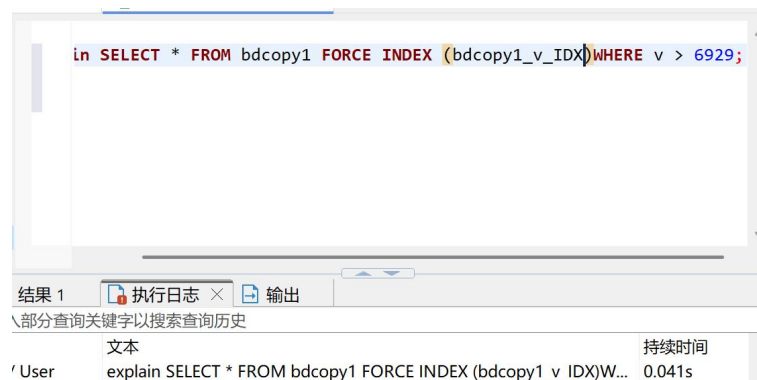
Bdcopy1, B+树, 等值查询, 基于键值排序, 从根节点根据键值遍历找到匹配项



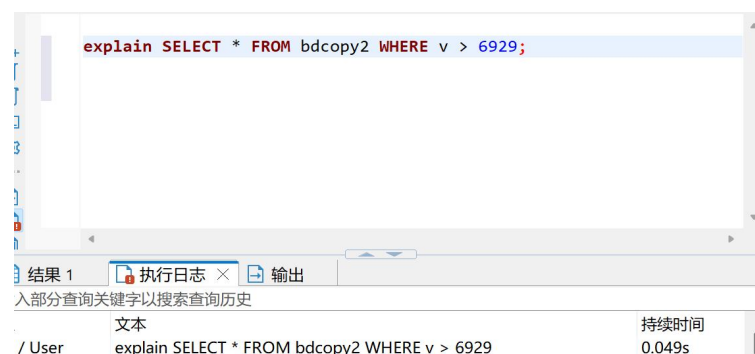
Bdcopy2, 没有建索引, 等值查询, 扫描整个表查找匹配的项目



Bdcopy1, B+树, 范围查询, 基于键值排序, 定位到起始叶子节点后顺序扫描链表, 时间优势小, 因此直接查询的时候会自动选择不使用索引

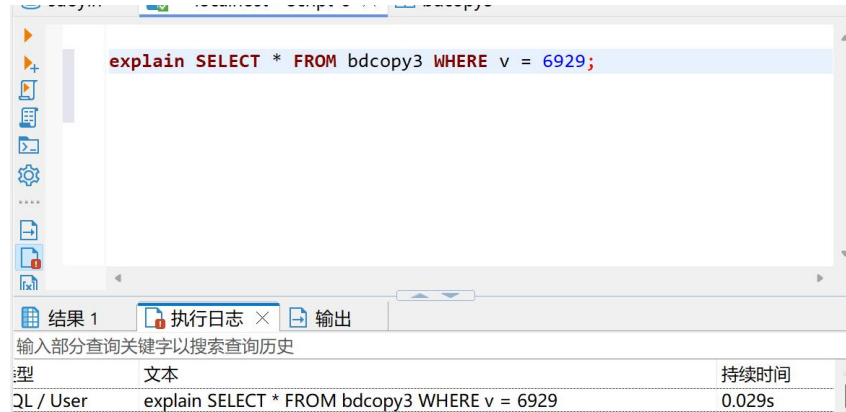


Bdcopy2, 没有建索引, 范围查询, 扫描整个表查找匹配的项目



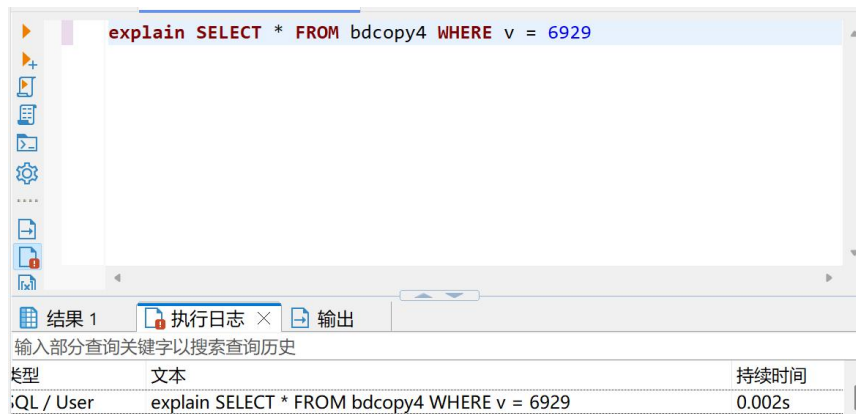
Q2

在 bdtable3 上建立的实则为 B+树索引，等值查询和 bdtable1 效果类似



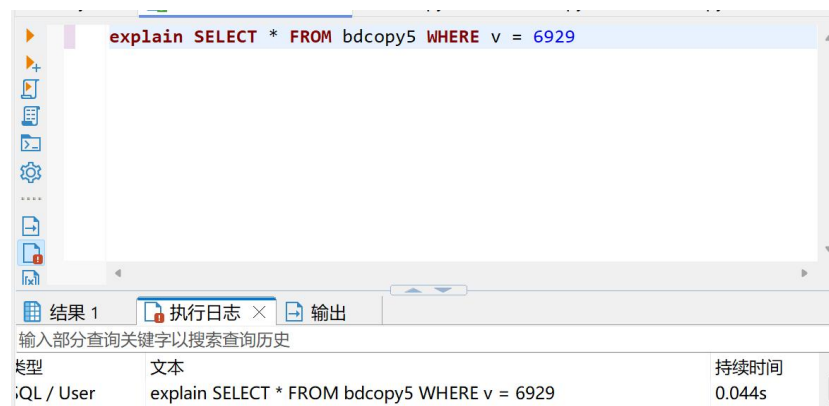
类型	文本	持续时间
QL / User	explain SELECT * FROM bdcopy3 WHERE v = 6929	0.029s

Bdtable4, 散列索引，等值查询，将键值映射的哈希值进行桶排序，等值查询效率高



类型	文本	持续时间
QL / User	explain SELECT * FROM bdcopy4 WHERE v = 6929	0.002s

Bdtable5, 无索引，等值查询，全表范围扫描



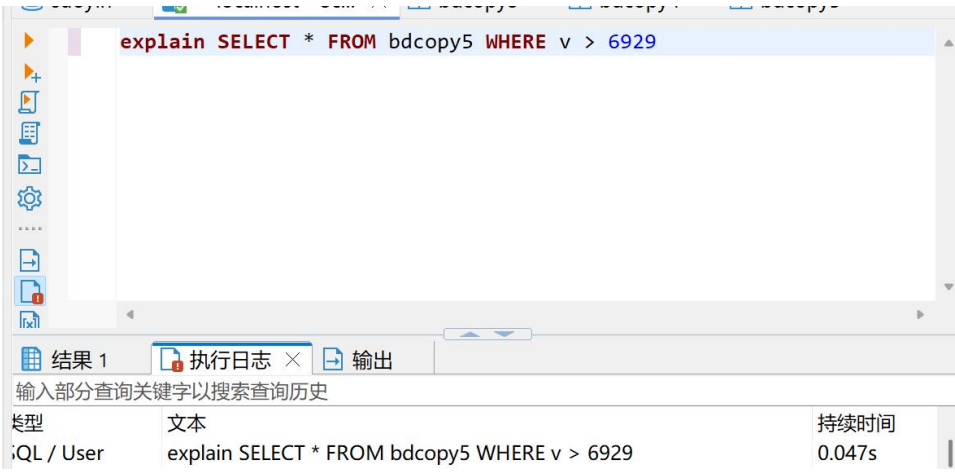
类型	文本	持续时间
QL / User	explain SELECT * FROM bdcopy5 WHERE v = 6929	0.044s

Bdtable4，散列索引，范围查询，不支持，仍旧使用没有索引的全表扫描



类型	文本	持续时间
SQL / User	explain SELECT * FROM bdcopy4 WHERE v > 6929	0.048s

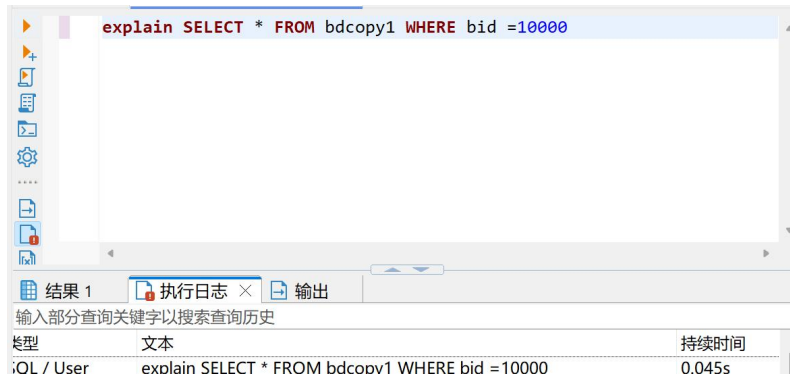
Bdtable5，无索引，范围查询，全表范围扫描



类型	文本	持续时间
SQL / User	explain SELECT * FROM bdcopy5 WHERE v > 6929	0.047s

Q3

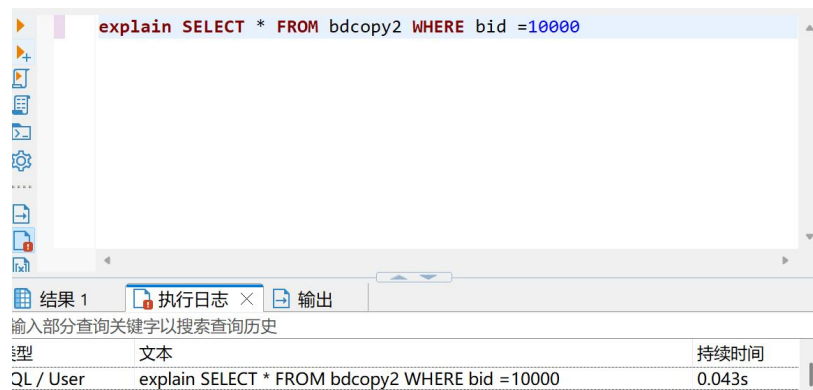
Bdcopy1, B+树, 等值查询, 基于键值排序, 从根节点根据键值遍历找到匹配项



The screenshot shows a SQL query window with the text: `explain SELECT * FROM bdcopy1 WHERE bid = 10000`. Below the query, the execution plan is displayed in a table format. The table has two columns: 'type' and 'duration'. The first row shows 'type' as 'Text' and 'duration' as '0.045s'. The second row shows 'type' as 'SQL / User' and 'duration' as '0.045s'.

type	duration
Text	0.045s
SQL / User	0.045s

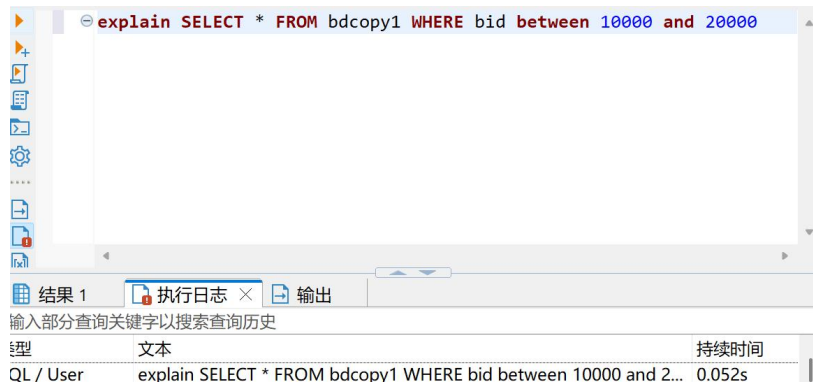
Bdtable2, 聚簇索引, 等值查询, 将数据和对应 B+树叶子节点物理上存储在一起, IO 较少



The screenshot shows a SQL query window with the text: `explain SELECT * FROM bdcopy2 WHERE bid = 10000`. Below the query, the execution plan is displayed in a table format. The table has two columns: 'type' and 'duration'. The first row shows 'type' as 'Text' and 'duration' as '0.043s'. The second row shows 'type' as 'QL / User' and 'duration' as '0.043s'.

type	duration
Text	0.043s
QL / User	0.043s

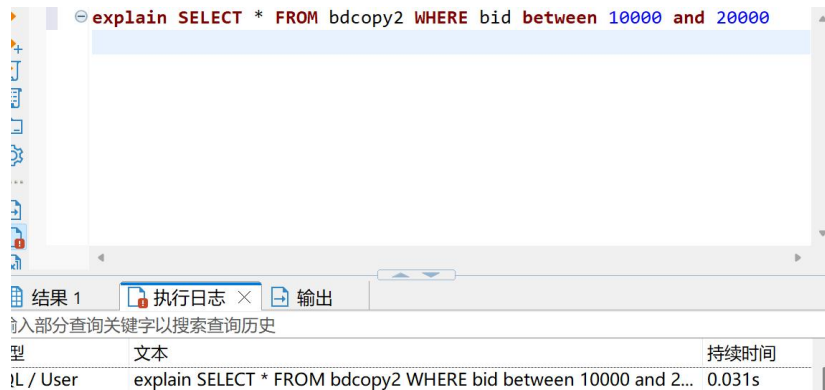
Bdcopy1, B+树, 范围查询, 基于键值排序, 定位到起始叶子节点后顺序扫描链表



The screenshot shows a SQL query window with the text: `explain SELECT * FROM bdcopy1 WHERE bid between 10000 and 20000`. Below the query, the execution plan is displayed in a table format. The table has two columns: 'type' and 'duration'. The first row shows 'type' as 'Text' and 'duration' as '0.052s'. The second row shows 'type' as 'QL / User' and 'duration' as '0.052s'.

type	duration
Text	0.052s
QL / User	0.052s

Bdtable2, 聚簇索引, 范围查询, 数据物理上连续存储, 顺序读取即可, 效率高

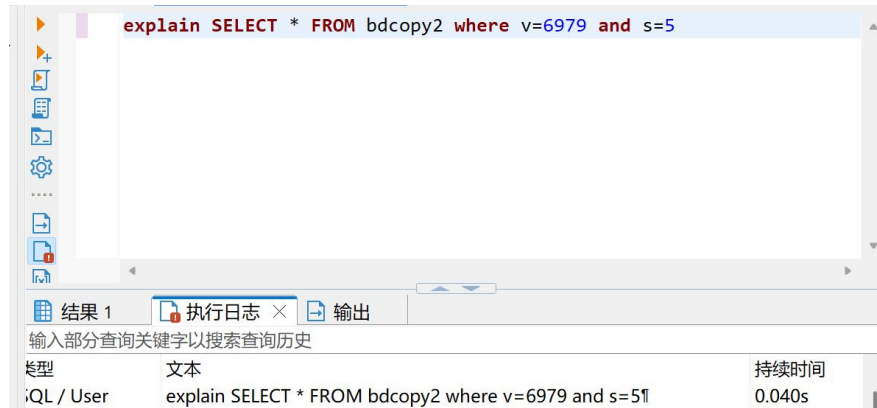


The screenshot shows a SQL query window with the text: `explain SELECT * FROM bdcopy2 WHERE bid between 10000 and 20000`. Below the query, the execution plan is displayed in a table format. The table has two columns: 'type' and 'duration'. The first row shows 'type' as 'Text' and 'duration' as '0.031s'. The second row shows 'type' as 'QL / User' and 'duration' as '0.031s'.

type	duration
Text	0.031s
QL / User	0.031s

Q4

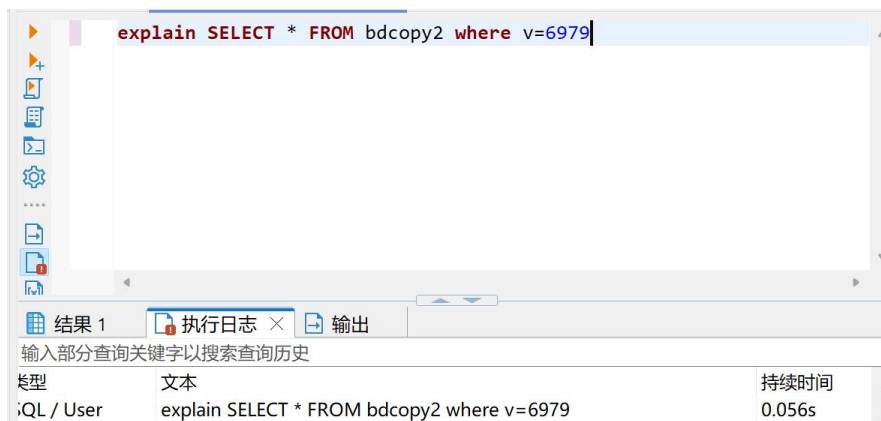
Bdtable2, 查询 v 和 s, 使用联合索引, 效率高



```
explain SELECT * FROM bdcopy2 where v=6979 and s=5
```

结果 1	执行日志	输出
输入部分查询关键字以搜索查询历史		
类型	文本	持续时间
SQL / User	explain SELECT * FROM bdcopy2 where v=6979 and s=5	0.040s

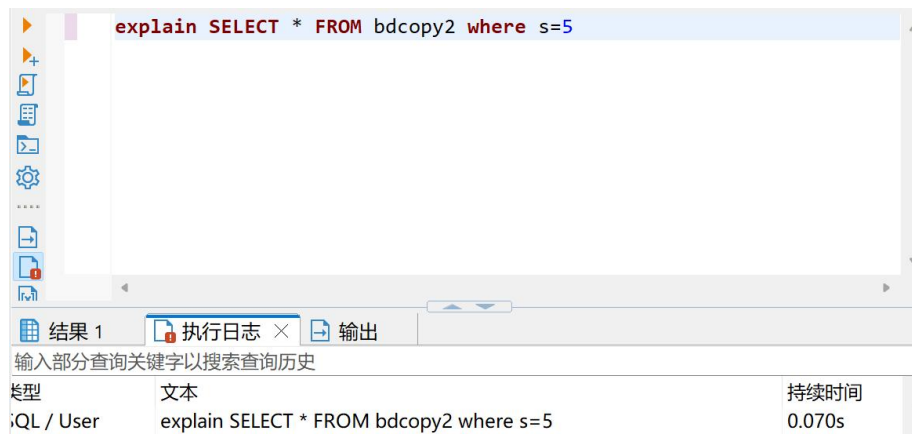
Bdtable2, 查询 v, 满足最左前缀原则, 使用联合索引, 效率高



```
explain SELECT * FROM bdcopy2 where v=6979
```

结果 1	执行日志	输出
输入部分查询关键字以搜索查询历史		
类型	文本	持续时间
SQL / User	explain SELECT * FROM bdcopy2 where v=6979	0.056s

Bdtable2, 查询 s, 不满足最左前缀原则, 不使用联合索引, 效率低

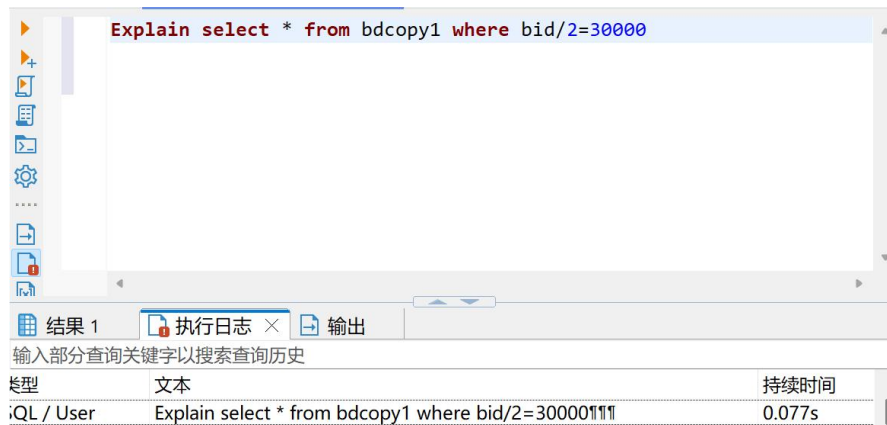


```
explain SELECT * FROM bdcopy2 where s=5
```

结果 1	执行日志	输出
输入部分查询关键字以搜索查询历史		
类型	文本	持续时间
SQL / User	explain SELECT * FROM bdcopy2 where s=5	0.070s

Q5

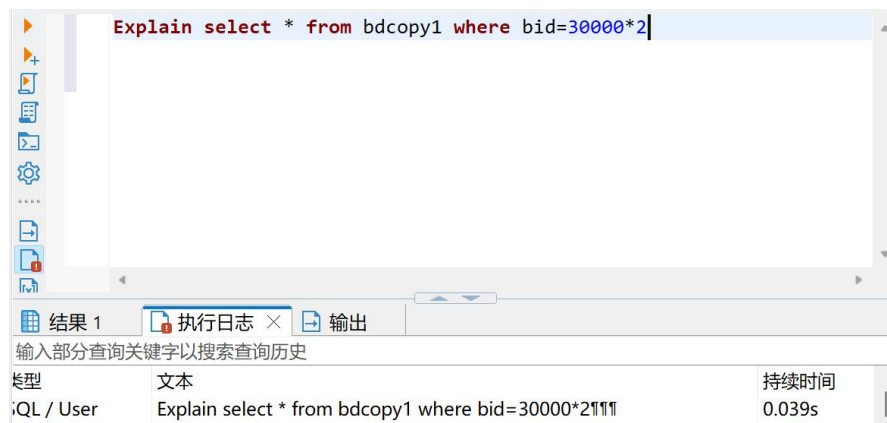
Bid/2=30000，左侧有计算式，索引不生效



The screenshot shows a database query tool interface. The top pane contains the SQL query: `Explain select * from bdcopy1 where bid/2=30000`. The bottom pane shows the execution results in a table format.

类型	文本	持续时间
SQL / User	Explain select * from bdcopy1 where bid/2=30000	0.077s

Bid=30000*2，左侧没有计算式，索引生效



The screenshot shows a database query tool interface. The top pane contains the SQL query: `Explain select * from bdcopy1 where bid=30000*2`. The bottom pane shows the execution results in a table format.

类型	文本	持续时间
SQL / User	Explain select * from bdcopy1 where bid=30000*2	0.039s